

Problem 1

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Problem 1 Evaluate the sum.

$$(a) \sum_{k=1}^n 5$$

Explanation. $\sum_{k=1}^n 5 = n \cdot 5$

$$(b) \sum_{k=1}^{10} 5$$

Explanation. Using the previous result for $n = 10$, we get $\sum_{k=1}^{10} 5 = 10 \cdot 5 =$

50

or

$$\sum_{k=1}^{10} 5 = 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 = 50$$

$$(c) \sum_{k=1}^n 8k$$

Explanation. $\sum_{k=1}^n 8k = 8 \sum_{k=1}^n k = 8 \frac{n(n+1)}{2} = 4n(n+1)$

$$(d) \sum_{k=1}^4 8k$$

Explanation. Using the previous result for $n = 4$, we get $\sum_{k=1}^4 8k =$

$$4(4)(4+1) = 80$$

or

$$\sum_{k=1}^4 8k = 8(1) + 8(2) + 8(3) + 8(4) = 8 + 16 + 24 + 32 = 80$$

$$(e) \sum_{k=1}^n (6k^2 - 8k - 5)$$

Explanation. $\sum_{k=1}^n (6k^2 - 8k - 5) = \sum_{k=1}^n 6k^2 + \sum_{k=1}^n (-8k) + \sum_{k=1}^n (-5) =$

$$= 6 \sum_{k=1}^n k^2 - 8 \sum_{k=1}^n k - \sum_{k=1}^n 5 =$$

Now we use the formulas.

$$= 6 \frac{n(n+1)(2n+1)}{6} - 8 \frac{n(n+1)}{2} - \sum_{k=1}^n 5 = n(n+1)(2n+1) - 4n(n+1) -$$

$$5n =$$

$$= 2n^3 - n^2 - 8n$$

$$(f) \sum_{k=0}^2 \cos\left(\frac{\pi}{2}k\right)$$

Explanation.

$$\sum_{k=0}^2 \cos\left(\frac{\pi}{2}k\right) = \cos\left(\frac{\pi}{2}(0)\right) + \cos\left(\frac{\pi}{2}(1)\right) + \cos\left(\frac{\pi}{2}(2)\right)$$

$$= \cos(0) + \cos\left(\frac{\pi}{2}\right) + \cos(\pi)$$

$$= 1 + 0 + -1 = 0.$$